Efficient Image Clustering using SIFT + CNN Ensemble Features

Caltech-101 Subset — BoVW + PCA — MobileNetV2 — Fusion Strategies

Zakariya Yahya Ally

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Problem & Goal

- **Task:** Unsupervised image clustering (group similar images without using labels for training).
- **Goal:** Compare clustering quality using:
  - Handcrafted local features: **SIFT** $\rightarrow$ **BoVW** $\rightarrow$ **PCA**
  - Deep semantic features: **MobileNetV2** (ImageNet pretrained)
  - **Fusion** beyond concatenation (averaging, weighted, non-linear ops)
- **Evaluation:** Silhouette (unsupervised), ARI & NMI (label-based reporting).

Dataset & Subset Used

Dataset: Caltech-101

- Total categories available: **102**
- Randomly selected classes (10):
  - `dolphin, minaret, trilobite, lamp, water_lilly, panda,`
  - `wheelchair, buddha, crocodile, ceiling_fan`
- Total images in subset: **604** (up to 50/class, depends on availability)

Why this subset is realistic:

- Mix of animals, objects, structures $\Rightarrow$ diverse shapes/textures.
- High intra-class variation and inter-class similarity $\Rightarrow$ challenging clustering.

SIFT $\rightarrow$ BoVW $\rightarrow$ PCA (Handcrafted Pipeline)

SIFT descriptors

- Each keypoint descriptor: **128-D**
- Total descriptors extracted (all images): **257,847** $\times$ 128

BoVW (Bag of Visual Words)

- Learn vocabulary using K-Means on descriptors: $K = 200$
- Each image becomes a histogram:

$$\mathbf{h}_i = [h_1, \ldots, h_{200}] \in \mathbb{R}^{200}$$

- BoVW feature matrix: **(604 $\times$ 200)**

PCA (Dimensionality reduction)

$$\mathbf{Z} = \mathbf{XW} \quad ; \quad 200 \rightarrow 50$$

CNN Feature Extraction (Deep Semantic Features)

Model: MobileNetV2 (ImageNet pretrained)

- Input size: **224×224**
- Pooling: **Global Average Pooling**
- Feature per image:

$$\mathbf{f}_{CNN} \in \mathbb{R}^{1280}$$

- CNN feature matrix: **(604 × 1280)**

Why CNN features help clustering:

- Encode higher-level semantics: shape, object parts, global structure.
- Stronger category separation than texture-only handcrafted descriptors.

Clustering Algorithm & Metrics

K-Means clustering with number of clusters:

$$k = \#\text{classes} = 10$$

Objective:

$$\min_{\{\mu_j\}_{j=1}^{k}} \sum_{i=1}^{N} \min_{j} \|\mathbf{x}_i - \mu_j\|^2$$

Evaluation metrics

- **Silhouette (unsupervised):**

$$s = \frac{b - a}{\max(a, b)} \in [-1, 1]$$

- **ARI** (agreement with true labels, adjusted for chance)
- **NMI** (normalized mutual information with true labels)

Fusion Strategies Tested (Beyond Concatenation)

Dimension alignment (CNN → 50D): $\mathbf{f}_{CNN}^{(50)} \leftarrow \mathrm{PCA}(\mathbf{f}_{CNN})$

Fusion methods (after scaling):

- **Averaging:** $\mathbf{f}_{avg} = \frac{1}{2}\left(\mathbf{f}_{SIFT} + \mathbf{f}_{CNN}^{(50)}\right)$
- **Weighted (CNN-dominant):** $\alpha = 0.8$ $\mathbf{f}_{w} = \alpha\, \mathbf{f}_{CNN}^{(50)} + (1-\alpha)\, \mathbf{f}_{SIFT}$
- **Power/Gamma:** $\gamma = 0.8$ $\mathbf{f}_{\gamma} = (\mathbf{f}_{SIFT})^{\gamma} \odot (\mathbf{f}_{CNN}^{(50)})^{1-\gamma}$
- **Log fusion:** $\mathbf{f}_{log} = \log(1 + \mathbf{f}_{SIFT}) + \log(1 + \mathbf{f}_{CNN}^{(50)})$

Also tested (caution): Exponential ($\beta = 2.0$), Geometric mean ($\epsilon = 10^{-9}$).

Results Summary (SIFT vs CNN vs Best Fusions)

Baselines:

Method	Silhouette	ARI	NMI
SIFT (BoVW+PCA)	-0.0122	0.0378	0.0946
CNN (MobileNetV2)	0.0756	0.7827	0.8941

Fusion (main approaches):

Fusion	Silhouette	ARI	NMI
Averaging	0.1911	0.8801	0.9401
Weighted ($\alpha = 0.8$)	0.1702	0.9100	0.9513
Power/Gamma ($\gamma = 0.8$)	0.1205	0.8640	0.9261
Log fusion	0.1552	0.7712	0.8880

Additional Fusion Results & Key Observations

Fusion	Silhouette	ARI	NMI
Exp fusion ($\beta = 2.0$)	0.8377	0.0012	0.0307
Geometric mean ($\epsilon = 10^{-9}$)	0.0246	0.2957	0.5746

Interpretation (real-world):

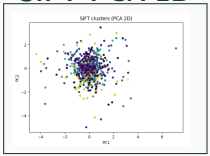
- **SIFT** fails: local textures are not enough for semantic grouping.
- **CNN** succeeds: pretrained deep features capture object semantics.
- **Averaging** is most stable: reduces noise and keeps shared structure.
- **Weighted fusion** gives best ARI/NMI: CNN dominates while SIFT adds complementary texture cues.
- **Exp fusion** is misleading: high geometric separation (Silhouette) but almost no semantic match (ARI/NMI ≈ 0) due to feature-space distortion.
- **Geometric mean** is too restrictive: keeps only agreement dims, suppresses

PCA Visualization (Qualitative Check)

How to read PCA plots:

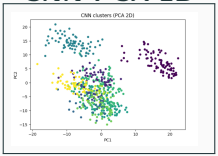
- Clear separated color groups ⇒ good clustering structure
- Heavy overlap ⇒ weak separability

SIFT PCA 2D



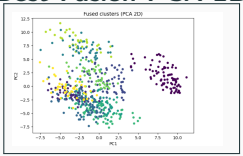
Overlap ⇒ weak semantic grouping

CNN PCA 2D



Better separation ⇒ semantic structure

Best Fusion PCA 2D



Strongest grouping (avg / weighted)

Comparison with Reference Papers (Context)

Why CNN features + fusion are expected to work (based on literature):

- **BoVW with SIFT**: classic pipeline for retrieval and grouping using a visual vocabulary; supports the SIFT+BoVW design.
- **Deep CNNs**: learn hierarchical features that outperform handcrafted features for visual understanding; explains the strong CNN baseline.
- **Off-the-shelf CNN features**: effective as generic descriptors across tasks (including grouping/retrieval); aligns with MobileNetV2 feature extraction.

Relation to our results:

- CNN features significantly outperform SIFT, and fusion improves performance further.
- This trend matches prior work showing that deep features provide stronger semantic separation.

Conclusion & Future Work

Conclusion (key points):

- **SIFT (BoVW+PCA)** alone produced weak clustering (low ARI/NMI).
- **CNN features** gave strong semantic clustering (high ARI/NMI).
- **Best improvements** came from fusion:
  - Averaging → best stability + strongest Silhouette among good methods.
  - Weighted fusion ($\alpha = 0.8$) → best semantic alignment (highest ARI/NMI).
- Aggressive non-linear transforms (e.g., exponential) can distort semantic structure.

Future work:

- Try additional clustering: Spectral, Agglomerative, DBSCAN.
- Use UMAP/t-SNE for visualization.
- Learn fusion weights automatically (trainable fusion / metric learning).

References

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